

Section I

10 marks

Attempt Questions 1 to 10

Allow approximately 15 minutes for this section

Mark your answers on the answer grid provided.

Questions

Marks

1. Let $\arg(z) = \frac{\pi}{5}$ for a certain complex number z . What is $\arg(z^7)$?

1

(A) $-\frac{7\pi}{5}$

(B) $-\frac{3\pi}{5}$

(C) $\frac{2\pi}{5}$

(D) $\frac{3\pi}{5}$

2. Which of following integrals uses the correct substitution for

1

$$\int_0^{\sqrt{3}} \frac{\ln(\tan^{-1}x)}{1+x^2} dx,$$

(A) $\int_0^{\sqrt{3}} \ln u \, du$

(B) $\int_0^{\frac{\pi}{6}} \frac{\ln u}{1+\tan^2 u} du$

(C) $\int_0^{\frac{\pi}{3}} \ln u \, du$

(D) $\int_0^{\frac{\pi}{3}} \frac{\ln u}{1+\tan^2 u} du$

3. What are the coordinates of the foci of the hyperbola $25y^2 - 16x^2 = 400$?

1

(A) $(0, \pm 4\sqrt{41})$

(B) $(\pm 4\sqrt{41}, 0)$

(C) $(0, \pm \sqrt{41})$

(D) $(\pm \sqrt{41}, 0)$

4. Given that $w^5 = 1$ and w is a complex number, what is the value of 1

$$1 + w + w^2 + w^3 + w^4 + w^5 ?$$

- (A) 1
(B) 0
(C) w
(D) $-w$

5. What is the eccentricity of the ellipse given by the equation: 1

$$\frac{x^2}{9} + \frac{y^2}{5} = 1$$

- (A) $\frac{2}{3}$
(B) $\frac{\sqrt{14}}{3}$
(C) $\frac{2}{\sqrt{5}}$
(D) $\sqrt{\frac{14}{5}}$

6. Given that α, β and γ are the roots of the polynomial $2x^3 + 4x - 5 = 0$. 1

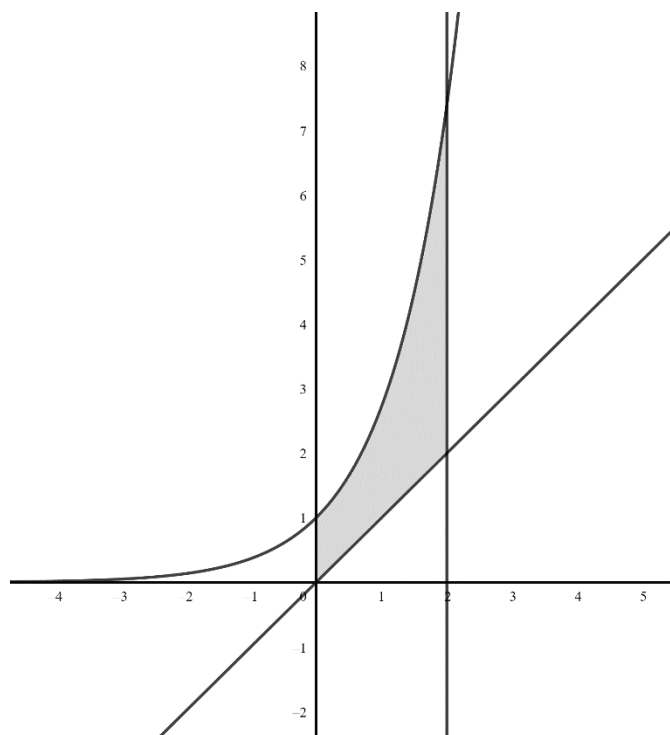
What is the value of $(\beta + \gamma - 3\alpha)(\alpha + \gamma - 3\beta)(\alpha + \beta - 3\gamma)$?

- (A) 32
(B) 160
(C) -32
(D) -160

7. Which of the following is an expression of $\int \frac{dx}{\sqrt{7-6x-x^2}}$? 1

- (A) $\sin^{-1}\left(\frac{x-3}{2}\right) + c$
(B) $\sin^{-1}\left(\frac{x-3}{4}\right) + c$
(C) $\sin^{-1}\left(\frac{x+3}{2}\right) + c$
(D) $\sin^{-1}\left(\frac{x+3}{4}\right) + c$

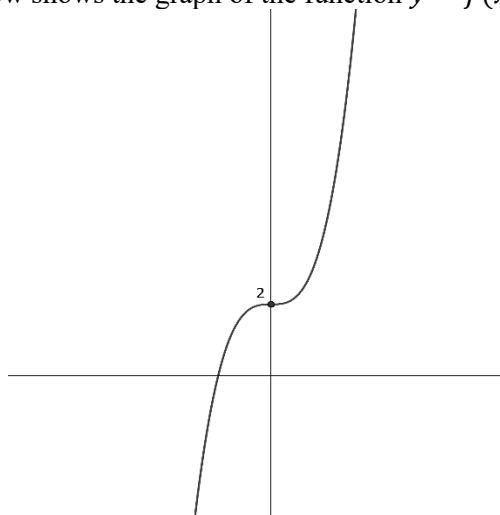
8. The region bounded by the curve $y = e^x$, the line $y = x$, the y-axis and the line $x = 2$ is rotated about the y-axis to form a solid. 1



Using method of cylindrical shells, which of the following is an expression for the volume V of the solid formed?

- (A) $2\pi \int_0^2 x(e^x - x) dx$
- (B) $2\pi \int_0^2 (x - e^x) dx$
- (C) $2\pi \int_0^2 (e^x - x) dx$
- (D) $2\pi \int_0^2 x(x - e^x) dx$
9. A particle of mass m falls vertically from rest under gravity in a medium in which the resistance to motion has magnitude $\frac{1}{20}mv^2$ where $v \text{ ms}^{-1}$ is the speed of the particle and $g = 9.8 \text{ ms}^{-2}$ is the acceleration due to gravity. What is the terminal velocity of the particle? 1
- (A) 9.8 ms^{-1}
- (B) 14 ms^{-1}
- (C) 19.6 ms^{-1}
- (D) 20 ms^{-1}

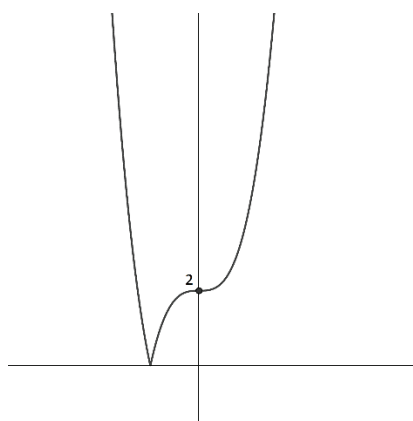
10. The diagram below shows the graph of the function $y = f(x)$:



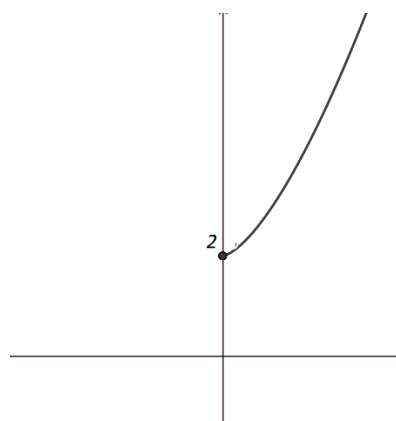
1

Which diagram represents the graph of $y = \sqrt{|f(x)|}$?

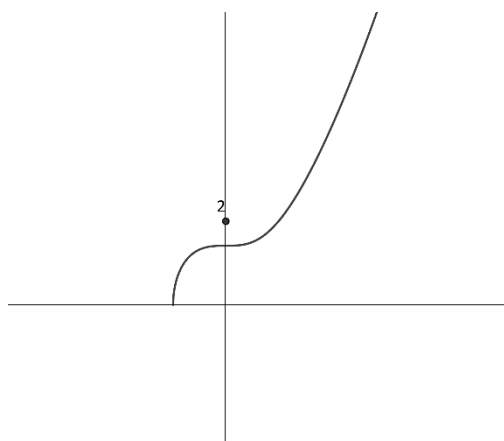
(A)



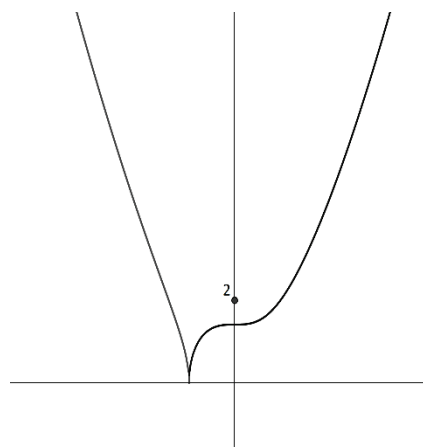
(C)



(B)



(D)



Section II

90 marks

Attempt Questions 11 to 16

Allow approximately 2 hours 45 minutes for this section.

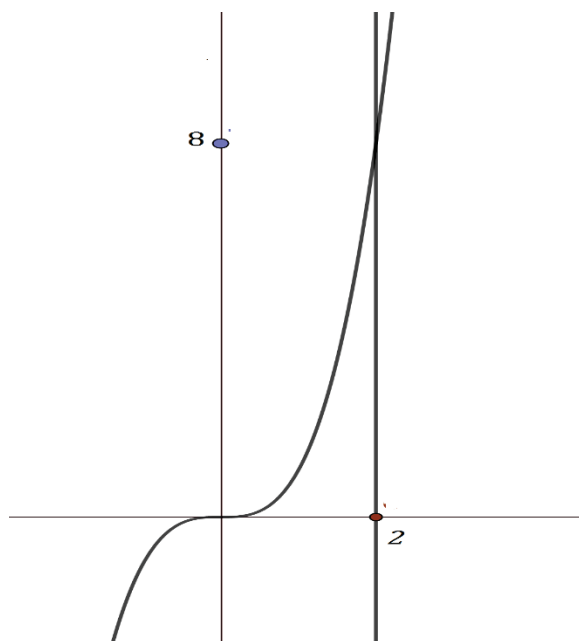
Write your answers in the writing booklets supplied. Additional writing booklets are available.
Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)	Marks
(a) For $z = 1 - i$, $w = 3 - 2i$, find:	
(i) $ z + w $	1
(ii) $z^2 - w^2$	1
(b) Find the exact value of:	
(i) $\int_2^3 \frac{x+1}{\sqrt{x^2+2x+5}} dx$	2
(ii) $\int_0^{\sqrt{2}} \sqrt{4-x^2} dx$	3
(c) Find the equation of the tangent to the curve $3x^2 - 2xy + y^3 = 1$ at the point $P(1, 1)$ to the curve.	3
(d) Sketch the region in the Argand diagram whose points satisfy:	3
$ z - 5i \leq 5$ and $ z + 5 > z - 5i $	
(e) Find:	2
$\int \frac{\ln x}{x^2} dx$	

Examination continues overleaf....

Question 12 (15 marks)

- (a) The region bounded by the curve $y = x^3$, the x -axis, $x = 0$ and $x = 2$ is rotated about the line $x = 2$. Find the volume of this solid using the method of slicing. 3



- (b) $z_1 = 1 + i$ and $z_2 = \sqrt{3} - i$
- (i) Find $z_1 \div z_2$ in the form $a + ib$ where a and b are real. 1
- (ii) Write z_1 and z_2 in modulus-argument form. 2
- (iii) Write $\cos \frac{5\pi}{12}$ as a surd by equating equivalent expressions for $z_1 \div z_2$. 1
- (c) (i) Find the values of A , B , C and D such that: 2

$$\frac{5x^3 - 3x^2 + 2x - 1}{x^4 + x^2} = \frac{A}{x} + \frac{B}{x^2} + \frac{Cx + D}{x^2 + 1}$$

- (ii) Hence integrate: 2

$$\int \frac{5x^3 - 3x^2 + 2x - 1}{x^4 + x^2} dx$$

- (f) (i) If α is a root of the polynomial $Q(x)$ with a multiplicity m , show that α is also a root of $Q'(x)$, with multiplicity $(m - 1)$. 1
- (ii) If the following polynomial $Q(x)$ has a triple root, factorise $Q(x)$ into its linear factors 3
- $$Q(x) = 2x^4 + 9x^3 + 6x^2 - 20x - 24.$$

Examination continues overleaf....

Question 13 (15 marks)

(a) Given that a , b and c are real positive numbers

(i) Prove that $a^2 + b^2 + c^2 \geq ab + bc + ac$. 2

(ii) Show that 2

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{9}{a + b + c}$$

(iii) Given that $a^2 + b^2 + c^2 = 9$, prove that 2

$$\frac{1}{1 + ab} + \frac{1}{1 + bc} + \frac{1}{1 + ac} \geq \frac{3}{4}$$

(b) If z and w are complex numbers such that $|z| = |w|$, show that 3

$$\frac{1}{2}(z + w) \cdot \frac{1}{2}(\overline{z + w}) + \frac{1}{2}(z - w) \cdot \frac{1}{2}(\overline{z - w}) = z\bar{z}.$$

(c) A particle of mass m kg is projected vertically upwards with a speed of $U \text{ ms}^{-1}$. At time t seconds the particle has vertical height x metres above the point of projection, speed $v \text{ ms}^{-1}$ and acceleration $a \text{ ms}^{-2}$. The particle moves under gravity in a medium where the resistance to motion has magnitude $\frac{m}{g}v^2$ Newtons where $g \text{ ms}^{-1}$ is the acceleration due to gravity.

(i) Show that $a = -\frac{1}{g}(g^2 + v^2)$. 1

(ii) Show that 3

$$v = g \left(\frac{U - g \tan t}{g + U \tan t} \right)$$

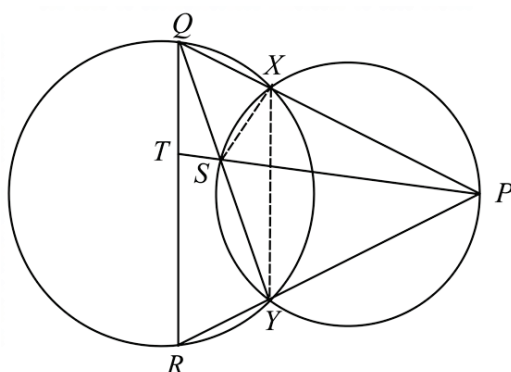
(iii) Find the time taken for the particle to reach its maximum height 1

(iv) Express x in terms of t . 1

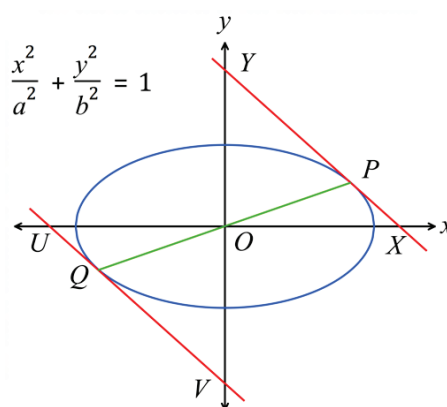
Examination continues overleaf....

Question 14 (15 marks)

- (a) The circles $XPYS$ and $XYRQ$ intersect at the points X and Y . PXQ , PYR , QSY , PST and QTR are straight lines.



- (i) Explain why $\angle STQ = \angle YRQ + \angle YPS$. 1
- (ii) Show that $\angle YRQ + \angle YPS + \angle SXQ = 180^\circ$. 2
- (iii) Prove that $STQX$ is a cyclic quadrilateral. 1
- (iv) Let $\angle QPY = \alpha$ and $\angle PQY = \beta$. Show that $\angle STQ = \alpha + \beta$ 3
- (a) $P(\text{acos}\theta, b\sin\theta)$ and $Q(\text{acos}\varphi, b\sin\varphi)$ are the end points of a diameter of the ellipse shown below.



Tangents to the ellipse at P , Q cut the x -axis at X , U respectively, and the y -axis at Y , V respectively.

- (i) Derive the expression for the gradient of the tangent to the ellipse and hence show that the tangent at P has the following equation.

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1 \quad 2$$

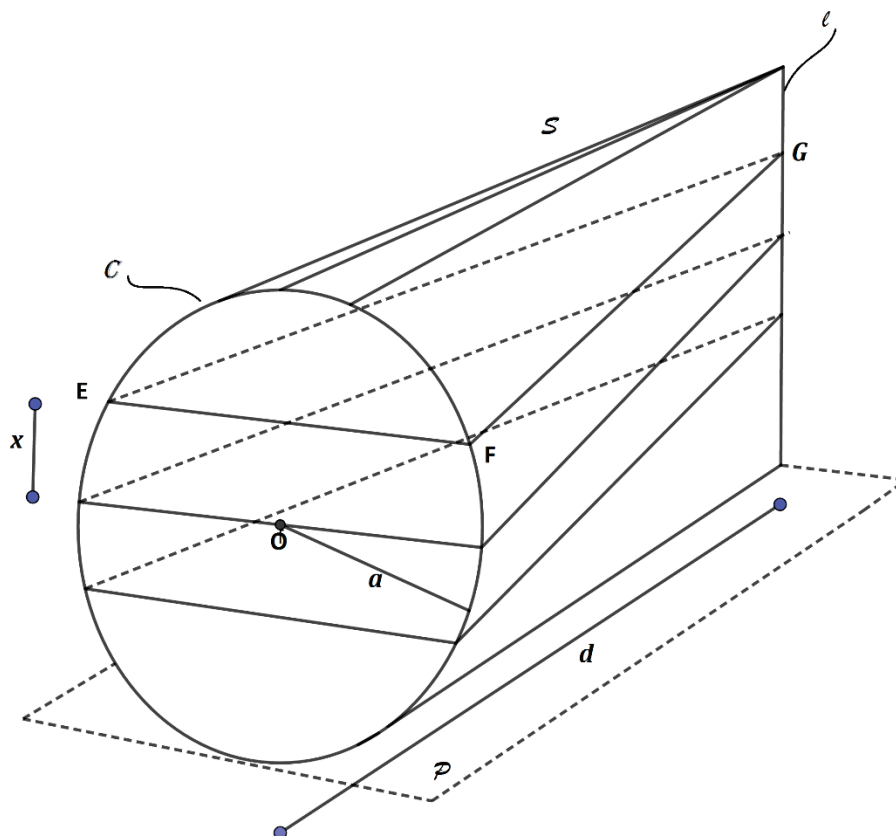
- (ii) Show that $\varphi = \theta \pm \pi$ 2
- (iii) What are the coordinates of X , Y , U , V in terms of a , b and θ ? 2
- (iv) Show that the area of $XYUV$ is 2

$$\frac{4ab}{|\sin 2\theta|}$$

Question 15 (15 marks)

- (a) The solid $\mathcal{S}$ is generated by moving a straight edge so that it is always parallel to a fixed plane P . It is constrained to pass through a circle C and line segment l .

The circle C , which forms a base for $\mathcal{S}$, has radius a and the line segment l is at a distance d from C . Both C and l are perpendicular to P . The perpendicular to C at its centre O bisects l .



- (i) Calculate the area of the triangular cross-section EFG which is parallel to P and distance x from the centre O of C . 2
- (ii) Calculate the volume of $\mathcal{S}$. 2
- (b) (i) Prove the identity: 2

$$\cos^3 A - \frac{3}{4} \cos A = \frac{1}{4} \cos 3A$$

- (iii) Show that $x = 2\sqrt{2}\cos A$ satisfies the cubic equation $x^3 - 6x + 2 = 0$ 2

given that $\cos 3A = -\frac{1}{2\sqrt{2}}$

- (c) For $n = 1, 2, 3, \dots$, S_n and T_n are two different sequences of positive integers.

$$\text{Given that } S_n = T_1 + T_2 + T_3 + T_4 \dots + T_n$$

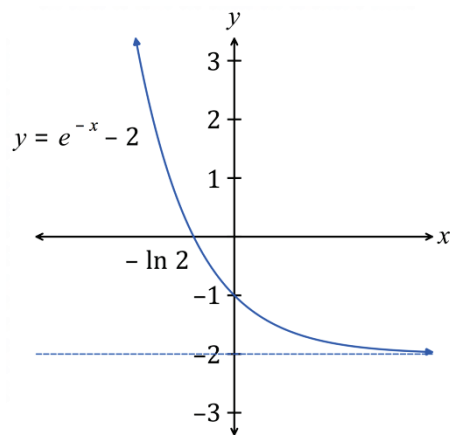
$$\text{Also } S_1 = 6, S_2 = 20 \text{ and } S_n = 6S_{n-1} - 8S_{n-2} \text{ for } n = 3, 4, 5 \dots$$

- (i) Prove by mathematical induction that $S_n = 4^n + 2^n$, $n = 1, 2, 3, \dots$ 4
- (ii) Hence or otherwise, find T_n , $n = 1, 2, 3, \dots$ in simplest form 3

Examination continues overleaf....

Question 16 (15 marks)

- (a) The graph of
- $f(x) = e^{-x} - 2$
- is shown below:



Draw separate one-third page sketches of the following functions. Indicate clearly any asymptotes and intercepts with the axes.

- (i) $y = |f(x)|$ 1
- (ii) $y = \{f(x)\}^2$ 1
- (iii) $y = \frac{1}{f(x)}$ 2
- (iv) $y = \ln f(x)$ 1
- (b) If a polynomial $P(x)$ is divided by $x^2 - u^2$, where $u \neq 0$, the remainder is $px + q$.

- (i) Show that 2

$$p = \frac{1}{2u} [P(u) - P(-u)] \text{ and}$$

$$q = \frac{1}{2} [P(u) + P(-u)]$$

- (ii) Find the remainder when $P(x) = x^n - u^n$, n is a positive integer, is divided by $x^2 - u^2$. 3

(Hint: Consider all possible cases for the value of n .)

- (c) For

$$I_n = \int_0^1 \frac{1}{(1+x^2)^n} dx \quad n = 1, 2, 3, \dots$$

- (i) Show that 3

$$I_{n+1} = \frac{2n-1}{2n} I_n + \frac{1}{n \times 2^{n+1}} \quad n = 1, 2, 3, \dots$$

- (ii) Hence evaluate 2

$$\int_0^1 \frac{1}{(1+x^2)^3} dx$$

Q1 $z = cis \frac{\pi}{5}$

$z^7 = cis \frac{7\pi}{5}$

$arg z = -\frac{3\pi}{5}$

[B]

Q2 let $\tan^{-1} x = u$

$\frac{1}{1+x^2} = \frac{du}{dx}$

$\frac{1}{1+x^2} = 0$

$\frac{1}{1+x^2} = 0$

$\frac{1}{1+x^2} = 0$

$u = \tan^{-1}(x) = \frac{\pi}{3}$

$\int \frac{\ln(\tan^{-1} x)}{1+x^2} dx$

$= \frac{\pi}{3} \int \ln u du$

[C]

Q3 $25y - 16x = 400$

$\frac{y^2}{16} - \frac{x^2}{25} = 1$

$a^2 = 25, b^2 = 16$

$a = 5, b = 4$

$\frac{25}{16} = e^2 - 1$

$e^2 = \frac{41}{16}$

$e = \pm \frac{\sqrt{41}}{4}$

for $i = (0 \pm be) = (0, \pm \sqrt{41})$

[C]

Q4 $\omega - 1 = 0$

$(\omega - 1)(1 + \omega + \omega^2 + \omega^3 + \omega^4) = 0$

$\Rightarrow 1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$

$\Rightarrow 1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$

Q5 $\frac{x^2}{a} + \frac{y^2}{5} = 1$

$b^2 = a^2(1 - e^2)$

$\frac{5}{a} = 1 - e^2$

$e^2 = 1 - \frac{5}{a} = \frac{4}{a}$

$e = \frac{2}{\sqrt{a}}$

[A]

Q6 $2x^2 + 4x - 5 = 0$

$\alpha + \beta + \gamma = 0, \alpha\beta\gamma = \frac{5}{2}$

$(\beta + \gamma - 3\alpha)(\alpha + \gamma - 3\beta)(\alpha + \beta - 3\gamma)$

$= (\alpha + \beta + \gamma - 4\alpha)(\alpha + \beta + \gamma - 4\beta)(\alpha + \beta + \gamma - 4\gamma)$

$= (-4\alpha)(-4\beta)(-4\gamma)$

$= -64(\alpha\beta\gamma) = -160$

[D]

Q7 $\int \frac{dx}{\sqrt{7-6x-x^2}}$

$= \int \frac{dx}{\sqrt{16-(x+3)^2}}$

$= \int \frac{dx}{\sqrt{16-(x+3)^2}}$

$= \sin^{-1}\left(\frac{x+3}{4}\right) + C$

[D]

Q8 radius of the shell = x

$\therefore$ Total $V = 2\pi \int_0^x (x^2 - x) dx$

[A]

Q9 Net force = mg - Resistance

$ma = mg - \frac{1}{20}mv^2$

$a = 9.8 - \frac{1}{20}v^2$

For Terminal velocity, $a = 0$

$v^2 = 9.8 \times 20$

$v = 19.6 \text{ m/s}$

[B]

Q10.

[D]

Q11

a) $z = 1 - i, w = 3 - 2i$

i) $z + w = 4 - 3i$

$|z + w| = \sqrt{16 + 9}$

$= 5$

ii) $z^2 - w^2$

$= (1 - i)^2 - (3 - 2i)^2$

$= (1 - 2i + i^2) - (9 - 12i + 4i^2)$

$= -2i - (5 - 12i)$

$= -5 + 10i$

b) i) $\int \frac{x+1}{\sqrt{x^2+2x+5}} dx$

let $u = x^2 + 2x + 5$

$\frac{du}{dx} = 2x + 2$

when $x = 2$

$u = 13$

when $x = 3$

$u = 20$

$\therefore \frac{1}{2} \int \frac{2(x+1)}{\sqrt{x^2+2x+5}} dx$

$= \int \frac{1}{\sqrt{u}} du$

$= \left[2\sqrt{u} \right]_{13}^{20}$

$= \sqrt{20} - \sqrt{13}$

ii) $I = \int \sqrt{4-x^2} dx$

let $x = 2 \sin \theta$

$\frac{dx}{d\theta} = 2 \cos \theta$

$\frac{dx}{d\theta} = 2 \cos \theta$

$\theta = 0$

$\theta = \frac{\pi}{4}$

$\theta = \frac{\pi}{4}$

$\therefore I = \int_0^{\frac{\pi}{4}} \sqrt{4-4\sin^2 \theta} \cdot 2 \cos \theta d\theta$

$= \int_0^{\frac{\pi}{4}} 4 \cos^2 \theta d\theta$

$= 2 \int_0^{\frac{\pi}{4}} (\cos 2\theta + 1) d\theta$

$= 2 \left[\frac{\sin 2\theta}{2} + \theta \right]_0^{\frac{\pi}{4}}$

$= 2 \left[\frac{1}{2} + \frac{\pi}{4} \right]$

$= \frac{1}{2} [2 + \pi]$

Q12 $3x^2 - 2xy + y^3 = 1$

$6x - (2x \frac{dy}{dx} + 2y) + 3y^2 \frac{dy}{dx} = 0$

$6x - 2y + (3y^2 - 2x) \frac{dy}{dx} = 0$

$\frac{dy}{dx} = \frac{2y-6x}{3y^2-2x}$

at $P(1,1)$

$\frac{dy}{dx} = \frac{2-6}{3-2} = -4$

Equation of the tangent at $P(1,1)$

$y - 1 = -4(x - 1)$

$4x + y - 5 = 0$

$$|z-5i| \leq 5$$

Circle with centre $5i$, radius 5

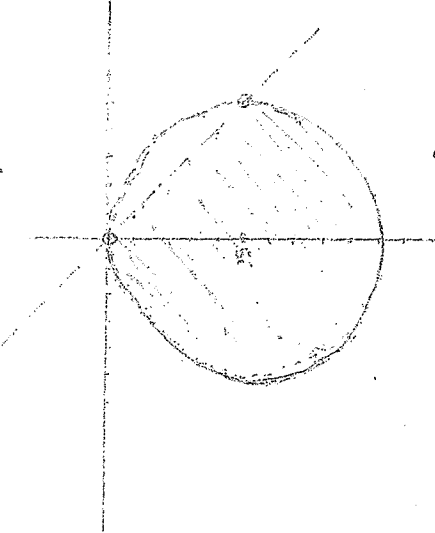
$$|z+5| > |z-5i|$$

$$|(x+5)+iy| > |x+(y-5)i|$$

$$\sqrt{(x+5)^2+y^2} > \sqrt{x^2+(y-5)^2}$$

$$10x+10y > 0$$

$$x+y > 0$$



$$I = \int \frac{\ln x}{x^2} dx$$

$$\text{let } u = \ln x, \quad v = -\frac{1}{x}$$

$$u' = \frac{1}{x}, \quad v' = \frac{1}{x^2}$$

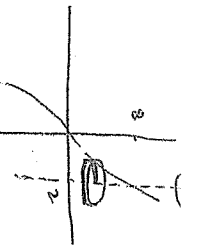
$$I = -\frac{1}{x} \cdot \ln x + \int \frac{1}{x^2} dx$$

$$= -\frac{\ln x}{x} - \frac{1}{x} + C$$

$$= -\frac{1}{x} (1 + \ln x) + C$$

Q12

a)



Radius of the slice = $2-x$
thickness = $8y$

Area of the cross-section of the slice = πr^2

$$= \pi (2-x)^2$$

$$= \pi (2-y^3)^2$$

Volume of the slice $\delta V = \pi (2-y^3)^2 \delta y$

Total volume

$$V = \lim_{\delta y \rightarrow 0} \sum \pi (2-y^3)^2 \delta y$$

$$= \pi \int_0^8 (2-y^{\frac{1}{3}})^2 dy$$

$$= \pi \int_0^8 (4 - 4y^{\frac{1}{3}} + y^{\frac{2}{3}}) dy$$

$$= \pi \left[4y - 4y^{\frac{4}{3}} \cdot \frac{3}{4} + y^{\frac{5}{3}} \cdot \frac{3}{5} \right]_0^8$$

$$= \pi \left[32 - 3(8)^{\frac{4}{3}} + \frac{3}{5} (8)^{\frac{5}{3}} \right]$$

$$= \pi \left[32 - 48 + \frac{3}{5} \times 32 \right]$$

$$V = \frac{16\pi}{5} \text{ units}^3$$

b)

$$Z_1 = 1+i$$

$$Z_2 = \sqrt{3}-i$$

$$i) \frac{Z_1}{Z_2} = \frac{(1+i)(\sqrt{3}+i)}{(\sqrt{3}-i)(\sqrt{3}+i)}$$

$$= \frac{\sqrt{3}+i+\sqrt{3}i-1}{4}$$

$$= \frac{\sqrt{3}-1}{4} + \frac{(\sqrt{3}+1)i}{4}$$

$$ii) Z_1 = 1+i$$

$$|Z_1| = \sqrt{2}$$

$$\arg Z_1 = \frac{\pi}{4}$$

$$Z_1 = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

$$Z_2 = \sqrt{3}-i$$

$$|Z_2| = 2$$

$$\arg Z_2 = -\frac{\pi}{6}$$

$$Z_2 = 2 \left(\cos \left(-\frac{\pi}{6} \right) + i \sin \left(-\frac{\pi}{6} \right) \right)$$

$$= 2 \left(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6} \right)$$

$$iii) \frac{Z_1}{Z_2} = \frac{\sqrt{2} \cos \frac{\pi}{4}}{2 \cos \frac{\pi}{6}}$$

$$= \frac{1}{\sqrt{2}} \cos \left(\frac{\pi}{4} + \frac{\pi}{6} \right)$$

$$= \frac{1}{\sqrt{2}} \cos \frac{5\pi}{12}$$

Comparing part (i) & (iii)

$$\frac{1}{\sqrt{2}} \cos \frac{5\pi}{12} = \frac{(\sqrt{3}-1)}{4}$$

$$\cos \frac{5\pi}{12} = \frac{\sqrt{2}(\sqrt{3}-1)}{4}$$

c)

$$i) \frac{5x^3 - 3x^2 + 2x - 1}{x^4 + x^2} = \frac{A}{x} + \frac{B}{x^2} + \frac{Cx+D}{x^2+1}$$

$$\Rightarrow A x(x^2+1) + B(x^2+1) + (Cx+D)(x^2)$$

$$= 5x^3 - 3x^2 + 2x - 1$$

$$= A x^3 + x + B x^2 + B + C x^3 + D x^2$$

$$= (A+C)x^3 + (B+D)x^2 + Ax + B$$

Comparing coefficients in (i) & (ii)

$$A+C = 5$$

$$B+D = -3$$

$$A = 2$$

$$B = -1$$

$$\Rightarrow C = 3$$

$$D = -2$$

$$ii) \text{ Hence } \int \frac{5x^3 - 3x^2 + 2x - 1}{x^4 + x^2} dx$$

$$= \int \frac{2}{x} - \frac{1}{x^2} + \frac{3x-2}{x^2+1} dx$$

$$= \int \frac{2}{x} - \frac{1}{x^2} + \frac{3x}{x^2+1} - \frac{2}{x^2+1} dx$$

$$= \int \frac{2}{x} - \frac{1}{x^2} + \frac{3}{2} \cdot \frac{2x}{x^2+1} - \frac{2}{x^2+1} dx$$

$$= 2 \ln x + \frac{1}{x} + \frac{3}{2} \ln(x^2+1) - 2 \tan^{-1} x$$

d

$$i) R(x) = (x-\alpha)^n \cdot P(x)$$

$$R'(x) = n(x-\alpha)^{n-1} \cdot P(x) + (x-\alpha)^n \cdot P'(x)$$

$$= (x-\alpha)^{n-1} [n \cdot P(x) + (x-\alpha) P'(x)]$$

$$R'(x) = (x-\alpha)^{n-1} R'(x)$$

$$\text{where } R(x) = n P(x) + (x-\alpha) P'(x)$$

Q(ii) α is a root of multiplicity 3

$$Q(x) = 2x^4 + 9x^3 + 6x^2 - 20x - 24$$

$$Q'(x) = 8x^3 + 27x^2 + 12x - 20$$

$$Q''(x) = 24x^2 + 54x + 12$$

α will be root of $Q'(x)$ with $m=2$

α root of $Q''(x)$ with $m=1$

$$\therefore Q'(x) = 24\alpha^2 + 54\alpha + 12 = 0$$

$$= 4\alpha^2 + 9\alpha + 2 = 0$$

$$(4\alpha + 1)(\alpha + 2) = 0$$

$$\Rightarrow \alpha = -\frac{1}{4}, \alpha = -2$$

Roots:

$$Q(-2) = 2(-2)^4 + 9(-2)^3 + 6(-2)^2 - 20(-2) - 24$$

$$= 32 - 72 + 24 + 40 - 24$$

$$= 0$$

$$\Rightarrow \alpha = -2 \text{ is a root of } m=3$$

$$Q(x) = (x+2)^3 P(x)$$

$$P(x) = Q(x) \div (x+2)^3$$

$$\begin{array}{r} 2x^2 \\ x+6x+12x+8 \quad 2x^4+9x^3+6x^2-20x-24 \\ \hline 2x^4+12x^3+24x^2+16x \\ \hline -3x^3-18x^2-36x-24 \\ \hline -3x^3-18x^2-36x-24 \\ \hline 0 \end{array}$$

$$\therefore Q(x) = (x+2)^3 (2x-3) \quad \checkmark$$

Q13

a) (i) $(a-b)^2 \geq 0$

$$a^2+b^2-2ab \geq 0$$

$$a^2+b^2 \geq 2ab \quad (1)$$

Similarly

$$b^2+c^2 \geq 2bc \quad (2)$$

$$a^2+c^2 \geq 2ac \quad (3)$$

Add (1), (2) & (3)

$$2(a^2+b^2+c^2) \geq 2(ab+bc+ac)$$

$$a^2+b^2+c^2 \geq ab+bc+ac$$

(ii)

From part (i)

$$(a-b)^2 \geq 0$$

$$(x-y)^2 \geq 0$$

$$x^2+y^2 \geq 2xy$$

$$\text{Let } x = \sqrt{\frac{a}{b}}, y = \sqrt{\frac{b}{a}}$$

$$\Rightarrow x^2+y^2 \geq 2xy$$

$$\frac{a}{b} + \frac{b}{a} \geq 2$$

Similarly

$$\frac{b}{c} + \frac{c}{b} \geq 2$$

$$\frac{a}{c} + \frac{c}{a} \geq 2$$

$$\text{Also } \begin{cases} \frac{a}{b} + \frac{b}{a} + \frac{a}{c} + \frac{c}{a} \geq 4 \\ \frac{b}{c} + \frac{c}{b} + \frac{b}{a} + \frac{a}{b} \geq 4 \\ \frac{a}{c} + \frac{c}{a} + \frac{b}{b} + \frac{c}{c} \geq 4 \end{cases}$$

Add $\frac{a}{b} + \frac{b}{a} + \frac{a}{c} + \frac{c}{a} + \frac{b}{c} + \frac{c}{b} + \frac{a}{a} + \frac{a}{a} + \frac{b}{b} + \frac{b}{b} + \frac{c}{c} + \frac{c}{c} \geq 12$

$$\frac{1}{a}(a+b+c) + \frac{1}{b}(a+b+c) + \frac{1}{c}(a+b+c) \geq 4$$

$$(a+b+c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \geq 9$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{9}{a+b+c}$$

(ii) From part (i)

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{9}{a+b+c}$$

$$\Rightarrow \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \geq \frac{9}{x+y+z}$$

$$\text{If } x = 1+a, y = 1+b, z = 1+c$$

$$\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c} \geq \frac{9}{1+a+b+c}$$

$$\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c} \geq \frac{9}{1+a+b+c}$$

$$\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c} \geq \frac{9}{1+a+b+c}$$

$$\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c} \geq \frac{9}{1+a+b+c}$$

$$\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c} \geq \frac{9}{1+a+b+c}$$

$$\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c} \geq \frac{9}{1+a+b+c}$$

$$\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c} \geq \frac{9}{1+a+b+c}$$

$$\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c} \geq \frac{9}{1+a+b+c}$$

$$|z| = |w|$$

$$|z|^2 = |w|^2$$

$$\therefore \frac{1}{4} [(2+u)(2+\bar{u}) + (2-u)(2-\bar{u})]$$

$$= \frac{1}{4} [(2+u)(2+\bar{u}) + (2-u)(2-\bar{u})]$$

$$= \frac{1}{4} [2\bar{z} + 2\bar{u} + 2\bar{u} + 2\bar{u}\bar{u} + 2\bar{z} - 2\bar{u} - 2\bar{u}\bar{u} + 2\bar{u}\bar{u}]$$

$$= \frac{1}{4} [2\bar{z}\bar{z} + 2\bar{u}\bar{u}]$$

$$= \frac{1}{4} [2\bar{z}\bar{z} + 2\bar{u}\bar{u}]$$

$$= \frac{1}{4} [2\bar{z}\bar{z} + 2\bar{u}\bar{u}]$$

W.C. mass = m kg

↑ Resistance
↓ motion

↓ Force due to gravity

(i) Net force = $-(mg + \text{Resistance})$

$$ma = -mg - \frac{m}{g} v^2$$

$$a = -g - \frac{v^2}{g}$$

$$a = -\frac{1}{g} (g^2 + v^2)$$

$$(ii) \therefore \frac{dv}{dt} = -\frac{1}{g} (g^2 + v^2)$$

$$\frac{dv}{v^2 + g^2} = -\frac{1}{g} dt$$

$$t = -g \int \frac{1}{g^2 + v^2} dv$$

$$t = -\tan^{-1}\left(\frac{v}{g}\right) + C$$

$$\text{When } t=0, v=0$$

$$\therefore -\tan^{-1}\left(\frac{v}{g}\right) + C = 0$$

$$C = \tan^{-1}\left(\frac{v}{g}\right)$$

$$\therefore t = -\tan^{-1}\left(\frac{v}{g}\right) + \tan^{-1}\left(\frac{v}{g}\right)$$

$$\text{Let } \tan^{-1}\left(\frac{v}{g}\right) = \alpha \therefore \frac{v}{g} = \tan \alpha$$

$$\therefore \tan^{-1}\left(\frac{v}{g}\right) = \beta$$

$$\therefore \tan^{-1}\left(\frac{v}{g}\right) = \beta$$

$$\therefore \tan^{-1}\left(\frac{v}{g}\right) = \beta$$

$$\therefore \tan^{-1}\left(\frac{v}{g}\right) = \beta$$

$$\therefore \tan^{-1}\left(\frac{v}{g}\right) = \beta$$

$$\therefore \tan^{-1}\left(\frac{v}{g}\right) = \beta$$

$$\therefore \tan^{-1}\left(\frac{v}{g}\right) = \beta$$

$$\therefore \tan^{-1}\left(\frac{v}{g}\right) = \beta$$

(Continued...)

$$\text{height} (g + v) = gU - gV$$

$$g^2 \text{height} + UV \text{height} + gV = gU$$

$$(U \text{height} + g)V = gU - g^2 \text{height}$$

$$V = \frac{g(U - g \text{height})}{g + U \text{height}}$$

ii) At max height, $V=0$

$$\frac{g(U - g \text{height})}{g + U \text{height}} = 0$$

$$U - g \text{height} = 0$$

$$\text{height} = \frac{U}{g}$$

$$t = \text{height} \left(\frac{U}{g} \right)$$

$$iv) \frac{dx}{dt} = \frac{g(U - g \text{height})}{g + U \text{height}}$$

$$= \frac{g(U - \frac{g \sin t}{\cos t})}{g + U \frac{\sin t}{\cos t}}$$

$$= \frac{g(U \cos t - g \sin t)}{g + U \frac{\sin t}{\cos t}}$$

$$= \frac{g(U \cos t - g \sin t)}{g \cos t + U \sin t}$$

$$x = g \int \frac{U \cos t - g \sin t}{g \cos t + U \sin t} dt$$

$$x = g \ln (g \cos t + U \sin t) + C$$

Q14 In ΔRPT

$$\angle PTE = \angle TRP + \angle RPT$$

[ext. angle is equal to sum of interior opposite angles]

$$\therefore \angle STE = \angle YRO + \angle YPS$$

ii) In quad. RYX

$$\angle TRY + \angle RYX = 180^\circ$$

(opposite angles of a cyclic quadrilateral)

$$\angle YRO + \angle RYX + \angle RYX = 180^\circ$$

$$\text{but } \angle RYX = \angle YPS \quad (\text{angles in the same segment})$$

$$\therefore \angle YRO + \angle YPS + \angle YRO = 180^\circ \quad (2)$$

iii) From part (i) $\angle YRO + \angle YPS = \angle STR$

from (2)

$$\angle STR + \angle SXO = 180^\circ$$

as $\angle STR$ & $\angle SXO$ are opp. angles of a quadrilateral/STX

$\Rightarrow \angle STR$ is a cyclic quad.

iv) $\angle XYS = \angle XPS$ & $\angle SXO = \angle XPO$
angles in the same segm.

$$\alpha = \angle QPY = \angle QPS + \angle XPS$$

$$= \angle XYS + \angle SXO = \angle XSO$$

$$\angle STR = \pi - \angle SXO$$

(opp. angles of a cyclic quadrilateral)

$$= \angle QRO + \angle XSO$$

(ext. angle of a triangle)

$$= \angle QRY + \angle QPY$$

(a b/c angle = sum of int. opposite angles)

$$= \alpha + \beta$$

$$b) \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$i) \frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{2x}{a^2} \times \frac{b^2}{2y}$$

$$= -\frac{b^2 x}{a^2 y}$$

$$\text{at } P(a \cos \theta, b \sin \theta)$$

$$\frac{dy}{dx} = -\frac{b^2 a \cos \theta}{a^2 b \sin \theta}$$

$$= -\frac{b \cos \theta}{a \sin \theta}$$

$\therefore$ Equation of tangent at P is:

$$y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$= a y \sin \theta - a b \sin^2 \theta = -b x \cos \theta + a b \cos^2 \theta$$

$$\Rightarrow b x \cos \theta + a y \sin \theta = ab$$

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

ii) $\angle UOQ = \angle XOP = \theta$ (vertically opposite angles)

$$\theta = 180^\circ + \theta \text{ i.e. } 180^\circ + \angle UOQ$$

$$= \pi + \theta$$

$$\alpha \phi = -(\pi - \angle UOQ)$$

$$= \theta - \pi$$

$$\therefore \phi = \theta \pm \pi$$

iii) Eq. of tangent at P

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

$$\text{at } X, y=0$$

$$\therefore X = \left(\frac{a}{\cos \theta}, 0 \right)$$

$$y \sin \theta = 1$$

$$\therefore y = \frac{b}{\sin \theta}$$

for tangent Q,

$$\text{Equation } \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

$$\frac{x \cos (\theta + \theta)}{a} + \frac{y \sin (\pi + \theta)}{b} = 1$$

$$-\frac{x \cos \theta}{a} + -\frac{y \sin \theta}{b} = 1$$

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = -1$$

$$\therefore \text{at } V, x=0$$

$$y = -\frac{b}{\sin \theta}$$

$$\text{at } V = x, y=0$$

$$\therefore x = -\frac{a}{\cos \theta}$$

iv) Diagonals of X Y U V:

$$YV \perp UX$$

& diagonals bisect each other

$\therefore$ XYUV is a rhombus

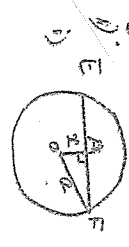
$$\therefore \text{area} = \frac{1}{2} \times (UX) (YV)$$

$$= \frac{1}{2} \cdot \frac{2a}{\cos \theta} \cdot \frac{2b}{\sin \theta}$$

$$= \frac{4ab}{2 \cos \theta \sin \theta}$$

$$= \frac{4ab}{\sin 2\theta}$$

Q15 -



ATF = $\sqrt{a^2 - x^2}$

$\therefore$ base of the triangle EFG
 $= 2\sqrt{a^2 - x^2}$

$\therefore$ area of the triangle
 $= \frac{1}{2} \cdot 2\sqrt{a^2 - x^2} \cdot a$
 $= a\sqrt{a^2 - x^2}$

(ii) $V = \int_{-a}^a a\sqrt{a^2 - x^2} dx$

$= a \int_{-a}^a \sqrt{a^2 - x^2} dx$
 $= a \int_{-a}^a \sqrt{a^2 - x^2} dx$

area of the semi circle
 $\pi \frac{1}{2} a^2$

$\therefore V = a \cdot \frac{1}{2} \pi a^2$
 $= \frac{\pi}{2} a^3$

b) $\cos 3A = \cos(2A + A)$
 $= \cos 2A \cos A - \sin 2A \sin A$
 $= (\cos^2 A - \sin^2 A) \cos A - 2 \sin A \cos A \cdot \sin A$
 $= \cos^3 A - \sin^2 A \cos A - 2 \sin^2 A \cos A$
 $= \cos^3 A - 3 \sin^2 A \cos A$
 $= \cos^3 A - 3(1 - \cos^2 A) \cos A$
 $= \cos^3 A - 3 \cos A + 3 \cos^3 A$
 $= 4 \cos^3 A - 3 \cos A$

(11) If $x = 2\sqrt{2} \cos A$ satisfies

the equation
 $x^3 - 6x + 2 = 0$

$\Rightarrow (2\sqrt{2} \cos A)^3 - 6(2\sqrt{2} \cos A) + 2 = 0$

Divide by $16\sqrt{2}$

$\cos^3 A - \frac{12\sqrt{2}}{16\sqrt{2}} \cos A + \frac{2}{16\sqrt{2}} = 0$

$\cos^3 A - \frac{3}{4} \cos A = -\frac{2}{16\sqrt{2}}$

$\frac{1}{4} \cos 3A = -\frac{2}{16\sqrt{2}}$

$\cos 3A = -\frac{8}{16\sqrt{2}}$

$\cos 3A = -\frac{1}{2\sqrt{2}}$

$\therefore x = 2\sqrt{2} \cos A$ satisfies the given equation.

c) $S_n = 4^n + 2^n$

Prove it true for $n=1, n=2$

$S_1 = 4 + 2 = 6$ True

$S_2 = 4^2 + 2^2 = 20$ True

Prove true for $n=1$ & $n=2$

Assume it to be true for:

$S_k = 4^k + 2^k$

To prove it true for $n=k+1$

$S_{k+1} = 4^{k+1} + 2^{k+1}$

Ans given

$S_n = 6S_{n-1} - 8S_{n-2}$

$\therefore S_{k+1} = 6S_k - 8S_{k-1}$

$= 6(4^k + 2^k) - 8(4^{k-1} + 2^{k-1})$

$= 6(4^k + 2^k) - 8(\frac{4^k}{4} + \frac{2^k}{2})$

$= 6(4^k + 2^k) - 2 \cdot 4^k - 4 \cdot 2^k$

$= (6-2)4^k + (6-4)2^k$

$= 4 \cdot 4^k + 2 \cdot 2^k$

$= 4^{k+1} + 2^{k+1}$

Hence proven true for $n=k+1$

$\therefore$ The statement is true for all $n \geq 1$, by principle of mathematical induction.

(ii) $S_n = T_1 + T_2 + \dots + T_n$

$\Rightarrow S_1 = T_1$

$S_2 = T_1 + T_2 = S_1 + T_2$

$S_3 = T_1 + T_2 + T_3 = S_2 + T_3$

$\Rightarrow S_n = S_{n-1} + T_n$

$\therefore T_n = S_n - S_{n-1}$

$= 4^n + 2^n - 4^{n-1} - 2^{n-1}$

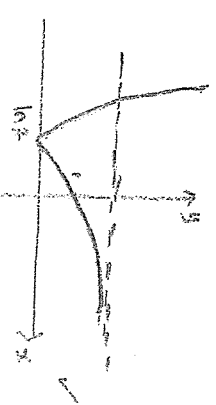
$= 4^n - 4^{n-1} + 2^n - 2^{n-1}$

$= (1 - \frac{1}{4})4^n + (1 - \frac{1}{2})2^n$

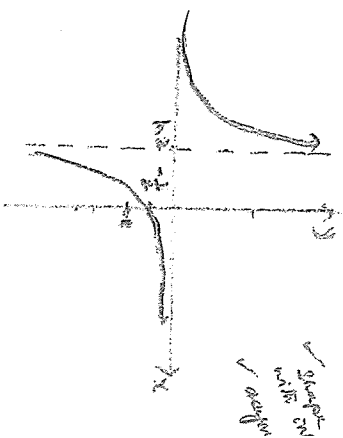
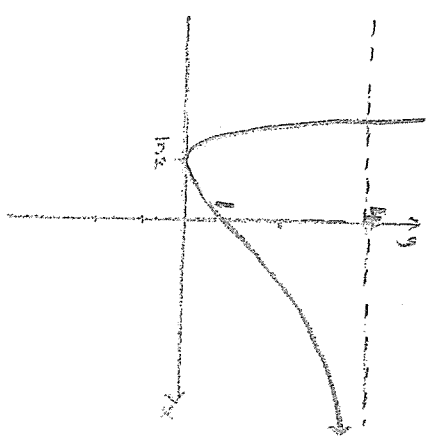
$= \frac{3}{4} \cdot 4^n + \frac{1}{2} \cdot 2^n$

$T_n = 2 \cdot 4^{n-1} + 2^{n-1}$

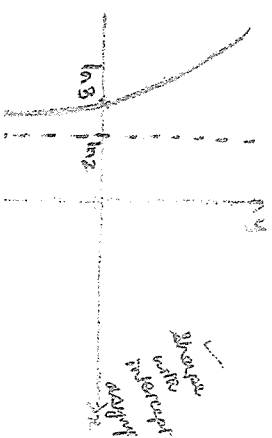
(12) (i)



area between the curves



area between the curves



area between the curves

2) $P(x)$ divided by $x^2 - u^2$

1) $P(x) = (x^2 - u^2) Q(x) + (px + q)$

$P(u) = pu + q$ (1)

$P(-u) = -pu + q$ (2)

(1) + (2)

$2q = P(u) + P(-u)$

$q = \frac{1}{2} [P(u) + P(-u)]$

2 (1) - (2)

$P(u) - P(-u) = 2pu$

$p = \frac{1}{2u} [P(u) - P(-u)]$

(ii) $P(x) = x^n - u^n$

If n is even

$P(u) = u^n - u^n = 0$

$u P(-u) = (-u)^n - u^n = 0$

$\Rightarrow P = \frac{1}{2u} [0 - 0] = 0$

$q = \frac{1}{2} [0] = 0$

$\therefore n$ is even
remainder = 0

If n is odd
 $P(u) = u^n - u^n = 0$

$P(-u) = (-u)^n - u^n = -2u^n$

$\therefore p = \frac{1}{2u} [2u^n] = u^{n-1}$

$q = \frac{1}{2} [-2u^n] = -u^n$

Remainder = $u^{n-1}x - u^n$
 $= u^{n-1} [x - u]$

(c) $u^n = \int_0^u \frac{1}{(1+x^2)^n} dx$

Let $u = \frac{1}{(1+x^2)^n}$ & $v = x$

$u' = \frac{-2nx}{(1+x^2)^{n+1}}$ & $v' = 1$

$\therefore I_n = \int \frac{x}{(1+x^2)^n} dx + 2n \int \frac{x^2}{(1+x^2)^{n+1}} dx$

$= \frac{1}{2n} + 2n \int (1+x^2-1) (1+x^2)^{-n-1} dx$

$= \frac{1}{2n} + 2n \int (1+x^2)^{-n} - (1+x^2)^{-n-1} dx$

$= \frac{1}{2n} + 2n I_n - 2n I_{n+1}$

$\therefore 2n I_{n+1} = \frac{1}{2n} + 2n I_n - I_n$

$I_{n+1} = \frac{1}{2n \cdot 2} + \frac{2n-1}{2n} I_n$

$= \frac{1}{n \cdot 2^{n+1}} + \frac{2n-1}{2n} I_n$

ii) $\int_0^1 \frac{1}{(1+x^2)^3} dx = I_3$ (n=2)

$I_3 = \frac{4-1}{4} I_2 + \frac{1}{2 \cdot 2^3}$

$= \frac{3}{4} I_2 + \frac{1}{16}$

$I_2 = \frac{1}{2} I_1 + \frac{1}{2^2}$

$= \frac{1}{2} I_1 + \frac{1}{4}$

$I_1 = \int_0^1 \frac{1}{1+x^2} dx = [\tan^{-1} x]_0^1 = \frac{\pi}{4}$

$\therefore I_2 = \frac{\pi}{8} + \frac{1}{4}$

$I_3 = \frac{3}{4} \left[\frac{\pi}{8} + \frac{1}{4} \right] + \frac{1}{16}$

$= \frac{3}{4} \left[\frac{3\pi}{8} + \frac{1}{4} \right] + \frac{1}{16} = \frac{9\pi}{32} + \frac{1}{8}$